

## Multiple Choice

1. **Answer:** A

*Options:* A) -28; B) 28; C) -7; D) 0; E) 7

*Note:* Correct option: A - -28

2. **Answer:** C

*Options:* A) 1200 volts/meter; B) 5 ohms; C) 1883 volts/meter; D) 940.5 volts/meter; E) 5 volts/meter

*Note:* Correct option: C - 1883 volts/meter

3. **Answer:** C

*Options:* A) 1; B) 2; C) 4; D) 8

*Note:* Correct option: C - 4

4. **Answer:** A

*Options:* A)  $A + B = |473| |375|$ ,  $A - B = |-1 -9 -1| |3 -23|$ ; B)  $A + B = |375| |475|$ ,  $A - B = |-2 -9 -1| |2 -32|$ ; C)  $A + B = |473| |475|$ ,  $A - B = |-1 -10 -1| |2 -31|$ ; D)  $A + B = |375| |475|$ ,  $A - B = |-2 -9 -2| |1 -31|$ ; E)  $A + B = |375| |474|$ ,  $A - B = |-1 -8 -1| |2 -32|$

*Note:* Correct option: A -  $A + B = |473| |375|$ ,  $A - B = |-1 -9 -1| |3 -23|$

5. **Answer:** A

*Options:* A) Process box.; B) State box.; C) Input box.; D) System box.; E) Data Box.

*Note:* Correct option: A - Process box.

## True / False

6. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

7. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: '0.015 joule'.

8. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

9. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: '500 MVA'.

10. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

## Long Form Response

11. **Answer:** 660 watts. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

*Note:* Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 29.7 kilowatts; 712,800 calories. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

*Note:* Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

*End of Answer Key*